

Elijah Zhao

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EDUCATION

University of British Columbia

Bachelor of Science, Major in Computer Science, Minor in Math

GPA: 4.1/4.33

Expected Graduation: May 2028

TECHNICAL EXPERIENCE

Software Engineer Intern

May 2026 – Present

Hootsuite

Vancouver, BC

- Built an **Apache Calcite**-based **JDBC** connector for Hootsuite Analytics, exposing data as relational tables and using filter pushdown to optimize Tableau and Power BI SQL queries; processed **5,000+** queries in its first week.
- Built the **first CI-gated Playwright E2E suite** for an MCP connector platform spanning three services; mocked flaky MCP and OAuth dependencies in a Docker Compose environment and covered **10+** failure paths.
- Built a Go middleware to track Analytics API usage, streaming events through **Kafka** into Mixpanel for analytics.

Software Engineer Intern

January 2026 – April 2026

Nexera Robotics (acquired by Locus Robotics)

Vancouver, BC

- Led redesign of a robotic pick-cycle pipeline in Python/C++, parallelizing ML inference and NumPy-based decision logic with **Ray Core** and optimizing critical paths in C++, improving throughput by **30% (7 to 10 picks/min)**.
- Built internal robotics tooling in Python/PostgreSQL interfacing with **Universal Robots** firmware, shipping **5+ features** including custom speed/pathing controls and performance tracking to identify system bottlenecks.
- Developed a [dataset curation pipeline](#) that reduced **1M+** unfiltered images to **7K** using **DINOv2** embeddings and **pgvector** similarity search for deduplication and sampling; stored images in **S3** and metadata/vectors in **RDS**.

Software Development Lead

August 2025 – Present

UBC BizTech

Vancouver, BC

- Leading **9** developers in building software and robotics platforms for UBC's largest tech club, serving **2,000+** users.
- Built a serverless backend with **AWS Lambda + Cognito** to manage authentication, payments, and event workflows; integrated **Stripe** webhooks to process **\$30K+** in membership payments and event confirmations.
- Designed **DynamoDB** access patterns with **GSIs** enabling queries across **9+** events and **900+** members.
- Shipped custom event products, including a [membership portal](#), [NFC networking system](#), and [mock stock exchange](#).

Undergraduate Teaching Assistant - CPSC 221

January 2026 – April 2026

University of British Columbia

Vancouver, BC

- Mentored **110+** students during lectures and weekly labs, teaching **data structures and algorithms** in **C++**.

PROJECTS

[NBA 3D Scene Reconstruction](#) | *Python, PyTorch, OpenCV, RF-DETR, SAM 2, Three.js*

Jul 2026 - Present

- Built a **computer vision** pipeline reconstructing monocular NBA clips as interactive 3D scenes in **Three.js**.
- Combined **RF-DETR/SAM 2** detections with **SigLIP** jersey embeddings to track and cluster players by team.
- Used court-keypoint homography and **MotionBERT** 3D pose lifting to reconstruct and animate player meshes.

[BizBot \(NwHacks 2026\)](#) | *Supabase, YOLOv8, FastAPI, Next.js, Google Gemini*

January 2026

- Won** Best Real-World Ready AI Product from **Block** among **630+** **hackers** by building an autonomous robot photographer with a **YOLOv8** vision loop to auto-trigger photo capture and publish an **AI-filtered** live gallery.

[Failure Cloud \(Berkeley AI Hackathon 2026\)](#) | *FastAPI, Next.js, PyBullet, NumPy*

May 2026

- Built a natural-language robot edge-case simulator generating portable **URDF/USD** scenarios for use across robotics platforms; demonstrated in **PyBullet** with RGB-D frames, point clouds, and failure evaluation.

[NBAsalary](#) | *Spring Boot, JPA, Python, Java, React, TailwindCSS*

August 2025 – September 2025

- Built a full-stack CRUD app with a **Spring Boot** backend, modeling relational entities and using **Spring Data JPA** for ORM and custom queries on **PostgreSQL**; integrated with a **React** frontend to visualize salary cap data.

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, JavaScript/TypeScript, SQL (PostgreSQL + pgvector)

Frameworks & ML: Spring Boot, React, Node.js, FastAPI, Ray, YOLOv8, OpenCV, RF-DETR, SAM 2, PyTorch

Cloud & Tools: AWS (Lambda/DynamoDB/EC2/Amplify/Cognito/API Gateway), Supabase, Docker, Git, Linux, Kafka